

PLAN — safs_seisol_v3_0_0_LSW (CSM depth-dependent stress + stress-tied LSW)

2026-06-15 — rev 1 (DESIGN)

Contents

0. Goal and scope	1
1. What is adopted from V2 unchanged vs changed	2
2. The CSM C1 k=2.39 stress field (what V3 adopts)	2
3. Stress injection: CSM field -> SeisSol via easi !ASAGI	3
4. Friction redesign (the core of V3)	4
5. Nucleation and friction parameters (hypocenter worked calc)	6
6. Implementation plan (files, once Section 8 is decided)	8
7. Validation gates (cluster; cannot be checked locally)	8
8. Open decisions (confirm before the build)	9
9. References (SAF-system / mechanics; verification status as of 2026-06-15)	9

- Date: 2026-06-15 (rev 1, DESIGN — no input files generated yet)
- Author target: SeisSol single-event dynamic rupture, FL=16 linear slip weakening
- Template: ../safs_seisol_v2_0_0_LSW/ (parameters.par, safs_fault.yaml, safs_initial_stress.yaml, safs_material_cvm.yaml/.nc, safs_mesh.puml.h5)
- New stress source: ../toolbox/on_fault_stress_projection_csm/ csm_yhsm2013_stress_on_safs_mesh_* — the C1 k=2.39, MUSCAL density, hydrostatic P_p, linear-sampled depth-dependent on-fault stress field.
- Constraint from the user: the rupture must propagate from the SE end (s ~ -18 km) all the way to the NW end (s ~ +266 km).
- Status: DESIGN FOR REVIEW. Building the input files is gated on the decisions in Section 8. None of this is verifiable locally — full propagation can only be confirmed on Frontera/Expanse (project policy: no local full-mesh runs).

0. Goal and scope

Reproduce the V2 LSW SeisSol case structure, but replace V2's single CONSTANT regional stress tensor with the depth-dependent, CSM-orientation stress field we just built, and redesign the linear-slip-weakening (LSW) friction so a single nucleated rupture runs the full SE -> NW length of the SAFS multi-strand fault.

SCOPE RULE (carried from the stress-projection plan): every parameter value must trace to SAF-system / southern-California observations or to setting-independent mechanics (Mohr-Coulomb friction, energy balance). No values imported from other earthquakes; Parkfield/SAFOD excluded.

1. What is adopted from V2 unchanged vs changed

UNCHANGED from safs_seisol_v2_0_0_LSW: - Mesh: safs_mesh.puml.h5 (same fault, 60,658 DR facets; UTM 11N; boundary codes 1=free surface, 3=DR fault, 5=absorbing). - Material: safs_material_cvm.yaml + safs_material_cvm.nc (MUSCAL CVM via easi !ASAGI). V3 reuses these verbatim. - parameters.par numerics: FL=16, numflux=godunov, Plasticity=0, CFL=0.5, ClusteredLTS=2, FixTimeStep=0.1, Format=10 (no volume output), refPointMethod=1 with XRef/YRef/ZRef=(0,-1,0), nucleation ramp s_0=0, t_0=1.0, fault &Elementwise output mask, energy + free-surface output. - Deep barrier concept: lock z in (-20000, -15000) m via mu_s -> 1e6. - Nucleation concept: time-ramped Gaussian over-stress Tnuc_s at the hypocenter (606971, 3707270, -4965.62).

CHANGED in V3: 1. Initial fault stress: from V2's single ConstantMap tensor to the SPATIALLY VARYING CSM C1 k=2.39 field (Section 3). 2. Friction: from V2's UNIFORM (mu_s=0.43, mu_d=0.12, d_c=1.8) to STRESS-TIED spatially varying mu_s and mu_d (Section 4) — this is forced by the new field's mu_app spread; uniform friction provably cannot meet the SE->NW constraint (Section 4.1).

2. The CSM C1 k=2.39 stress field (what V3 adopts)

Source artifact (this repo, regenerated 2026-06-15): toolbox/on_fault_stress_projection_csm/csm_yhsm2013_s + _summary.json. Built with: --magnitude-mode depth-R --closure ratio --k-ratio 2.39 --density-source muscal --pp-model hydrostatic --axes csm --sample linear.

Per-facet effective tractions (compression-positive, SEAS convention), fault-wide stats:

```
sigma_n_eff:  min 0.20   median 111.7   max 415.4   MPa   (grows with depth)
tau_mag:      min 0.06   median  29.2    max 160.2   MPa
mu_app=tau/sn: min 0.014 median  0.275   max  0.449
percentiles:  p75 0.340  p90 0.376  p95 0.397  p99 0.423
fraction mu_app > 0.40: 4.3 %   > 0.35: 21.9 %
hypocenter facet (s=0): sigma_n_eff 70.1, tau 20.8, mu_app 0.296, z -5054 m
```

ALONG-STRIKE mu_app profile (seismogenic band z = -12..-3 km, strike azimuth 314 deg, s measured from the hypocenter, NW positive; SE end at s ~ -18 km, NW end at s ~ +266 km):

s (km)	mu_app(med)	sigma_n_eff(med MPa)	note
-20..0	0.246	95.8	
0..20	0.352	81.0	hypocenter (good nucleation)
20..40	0.262	102.0	
40..60	0.142	120.4	*** PRIMARY GATE (p10 0.103)
60..80	0.246	125.0	

80..100	0.376	140.9	
100..120	0.268	118.2	
120..140	0.240	123.4	
140..160	0.309	109.2	
160..180	0.352	135.0	
180..200	0.231	184.9	
200..220	0.308	150.0	
220..240	0.181	169.4	*** SECOND GATE (p10 0.144)
240..260	0.333	174.5	

The two low- μ_{app} corridors ($s=40-60$ km, $\mu_{app} \sim 0.14$; $s=220-240$ km, $\mu_{app} \sim 0.18$) are the propagation gates. The $s=40-60$ km gate is the mechanical analog of the San Geronio restraining knot: the CSM stress is poorly oriented for slip on the local fault strike there, so the resolved shear-to-normal ratio collapses. This gate is DEEPER (lower μ_{app}) than V2's ~ 0.25 corridor where candidates B and C arrested.

3. Stress injection: CSM field -> SeisSol via easi !ASAGI

V2 fed SeisSol a single global Cartesian tensor (ConstantMap) and let SeisSol project it per facet (BaseDRInitializer rotateStressToFaultCS). The CSM field is NOT a single tensor; it is a spatially varying tensor field $\sigma_{eff}(x,y,z)$. SeisSol's supported mechanism for that is a gridded easi !ASAGI field (MFEM_to_SeisSol_mapping.md Section 2, line: "a gridded field via easi !ASAGI (HDF5)"), identical in form to the material safs_material_cvm.nc workflow.

KEY PROPERTY (why this is clean for stress): the CSM effective tensor is a genuine VOLUMETRIC field — its orientation is the CSM principal-axis field interpolated in (x,y) (depth-invariant), its shape ratio R is $CSM(x,y)$, and its magnitude is $Sv_{eff}(z)$ (MUSCAL lithostat - hydrostatic P_p) combined with the closure. So $\sigma_{eff}(x,y,z)$ can be evaluated on any 3D grid and SeisSol projects it per facet exactly as it did the V2 constant tensor.

3.1 New tool: CSM stress -> ASAGI nc

Add toolbox/on_fault_stress_projection_csm/csm_stress_to_asagi.py (sibling of project_csm_stress_to_vt reusing its tensor builder):

```
inputs : the CSM csv (orientations), --k-ratio 2.39, MUSCAL nc (density),
        pp-model hydrostatic (same knobs as the projection deliverable)
grid   : rectilinear UTM 11N covering the fault bounding box with margin
        x: ~360..625 km, y: ~3690..3840 km, z: -17000..0 m
        spacing: dz = 250 m (match material nc); dx=dy = 500 m
        (the fault facets are ~430 m; 500 m resolves the along-strike
         mu_app gate structure; finer only needed near junctions, see
         Section 4.4)
compute: at every grid node evaluate sigma_eff(x,y,z) (e,n,u = x,y,z),
        then NEGATE to SeisSol compression-NEGATIVE convention.
        P_p is already removed (effective tensor) -> baked-in, exact under
```

```

    projection (isotropic part projects to sigma_n only).
output : COARDS NetCDF, compound variable stress{s_xx,s_yy,s_zz,s_xy,
    s_yz,s_xz} laid out data(z,y,x), Pa (mirror convert_cvm_to_asagi)
selfcheck: trilinear sample of the written nc at the 60,658 fault facet
    centroids must reproduce the projection VTU's per-facet
    sigma_n_eff / tau within tolerance (the projection VTU is ground
    truth). FAIL the converter if not.

```

3.2 safs_initial_stress.yaml (V3)

Replace V2's ConstantMap with:

```

[s_xx,s_yy,s_zz,s_xy,s_yz,s_xz]: !ASAGI
  file: safs_stress_csm.nc
  parameters: [s_xx,s_yy,s_zz,s_xy,s_yz,s_xz]
  var: data
  interpolation: linear

```

(with a !ConstantMap fallback to V2's tensor for graceful degradation, following the material yaml pattern.)

4. Friction redesign (the core of V3)

4.1 Uniform LSW friction is INFEASIBLE for this field (proof)

To avoid spontaneous failure at $t=0$, the static friction must exceed the initial ratio everywhere: $\mu_s > \max(\mu_{app}) = 0.449$, so $\mu_s \sim 0.46$. With a uniform μ_s , the strength excess at the primary gate is

$$SE_{gate} = (\mu_s - \mu_{app}) * \sigma_n = (0.46 - 0.142) * 120 = 38.3 \text{ MPa}$$

The largest stress drop physically available there (even at $\mu_d = 0$) is

$$drop_{max} = (\mu_{app} - 0) * \sigma_n = 0.142 * 120 = 17.0 \text{ MPa}$$

so the S-ratio $S = SE/drop \geq 38.3/17.0 = 2.25$ (and ≥ 3.6 at the $\mu_{app} \sim 0.10$ patches), far above the ~ 1.5 unbounded-growth threshold. No uniform (μ_s, μ_d) passes the gate; it would require a negative μ_d . The user's $\mu_d \sim 0.10$ was tested explicitly: at the gate it yields a 5 MPa drop against a 38 MPa strength excess ($S = 7.5$), and ZERO drop on the $\mu_{app} \sim 0.10$ patches. μ_d is not the binding knob — μ_s is. This is the same arrest mechanism that stopped V2 candidates B and C, now made worse by the deeper gate.

4.2 Stress-tied friction (the design)

Tie μ_s and μ_d to the local apparent friction so the strength excess and stress drop (hence S) are CONTROLLED along the entire fault:

```

mu_s(facet) = mu_app(facet) + delta_s
mu_d(facet) = max( mu_app(facet) - delta_d , mu_d_floor )
=> strength excess SE = delta_s * sigma_n      (uniform in mu)
    stress drop      dt = delta_d * sigma_n      (where unfloored)
    S = delta_s / delta_d                        (uniform, low)

```

RECOMMENDED values (Section 8 lists these as the decisions to confirm):

```

delta_s   = 0.05      (t=0 margin; SE = 5% of sigma_n; mu_s = mu_app+0.05)
delta_d   = 0.15      (dynamic drop = 15% of sigma_n where unfloored)
mu_d_floor= 0.00      (full weakening on the lowest-mu_app patches)
=> S = delta_s/delta_d = 0.33 everywhere (robust, well below 1)

```

Resulting ranges over the fault:

```

mu_s = mu_app + 0.05 : 0.06 .. 0.50 (mu_s > mu_app by 0.05 => t=0 stable)
mu_d = max(mu_app-0.15, 0): 0 .. 0.30 (MEDIAN ~0.125 ~ the user's ~0.10)
primary gate (mu_app 0.142, sn 120): mu_s 0.192, mu_d 0 (floored),
    drop = 0.142*120 = 17 MPa, SE = 0.05*120 = 6 MPa, S = 0.35 -> PASSES
hypocenter (mu_app 0.296, sn 70): mu_s 0.346, mu_d 0.146,
    drop = 0.15*70 = 10.5 MPa, SE = 3.5 MPa, S = 0.33
highest mu_app 0.449: mu_s 0.499, mu_d 0.299 (no t=0 failure)

```

This is a standard heterogeneous-prestress technique: by referencing the friction to the prestress, the fault is everywhere uniformly (and weakly) super-critical once nucleated, so a single nucleation runs the full length. The low absolute μ_d (median ~ 0.125 , floor 0) is consistent with the SAF weak-fault / low-heat-flow evidence (Lachenbruch and Sass 1980, 1992; Hardebeck and Hauksson 2001; Townsend and Zoback 2004) — Section 9.

4.3 Cohesive-zone resolution (d_c)

Linear slip-weakening cohesive-zone length (Day et al. 2005 form):

```

Lambda = (9*pi/32) * mu_shear * d_c / ((mu_s - mu_d) * sigma_n)

```

With $(\mu_s - \mu_d) = \delta_s + \delta_d = 0.20$ (uniform by construction), $\mu_{\text{shear}} \sim 24$ GPa, $d_c = 2.0$ m:

```

sigma_n = 120 MPa (gate) : Lambda = 1473 m (~3 facet widths @ 430 m)
sigma_n = 300 MPa (deep) : Lambda = 589 m (~1.4 widths -> MARGINAL deep)
sigma_n = 70 MPa (hypo) : Lambda = 2526 m (well resolved)

```

RECOMMEND $d_c = 2.0$ m (V2 used 1.8 m). The deep facets ($\sigma_n > \sim 250$ MPa, $z < \sim -12$ km, near the locked barrier) are marginally resolved; mitigations if a convergence check fails: (a) raise d_c , or (b) tie d_c to σ_n so Λ is constant. Deferred until the first run shows whether the deep band matters (it is adjacent to the $\mu_s = 1e6$ barrier).

4.4 Implementation of stress-tied friction (the hard part)

$\mu_{app} = |\tau| / \sigma_n$ is a FAULT-SURFACE quantity — it depends on the fault normal, so unlike the stress tensor it is NOT a clean volumetric function of (x,y,z) and cannot be written as a Lua $f(x,y,z)$. It must be PRE-COMPUTED per facet (the projection tool already does, knowing each normal) and delivered to SeisSol's fault easi, which is evaluated at fault Gauss points (x,y,z) . Two mechanisms:

Option A (RECOMMENDED): rasterize per-facet μ_s , μ_d to a 3D ASAGI nc. The converter computes μ_{app} per facet, forms μ_s/μ_d per Section 4.2, and writes a grid where each node carries the value of the NEAREST fault facet (within a band; benign fill outside). SeisSol's linear ASAGI lookup at a fault point returns ~ that facet's value (neighbors vary smoothly under `--sample linear`). `safs_fault.yaml`: `[\mu_s] / [\mu_d] !ASAGI`. RISK: at multi-strand junctions (`strands < grid-cell` apart) nearest-facet is ambiguous; mitigate with `dx=dy=250` m near the fault or a signed band.

Option B: provide per-facet values keyed by an along-fault parameterization if SeisSol's fault input supports it (needs verification in the QuakeWorx SeisSol build; not assumed here).

The same rasterization can also emit fault-local tractions $Ts0/Td0/Pn0$ if we prefer per-facet stress over the volumetric tensor of Section 3 — that guarantees $\mu_s - \mu_{app} = \delta_s$ EXACTLY (Section 8 decision 2).

4.5 Deep barrier and shallow note

- Deep barrier: keep V2's lock, $\mu_s = 1e6$ for z in $(-20000, -15000)$ m, via the same Lua override layered on top of the ASAGI μ_s (or baked into the rasterized μ_s field).
- Free surface: NOTE the low- μ_{app} gate is at DEPTH ($s=40-60$ km, $z -12..-3$ km), NOT near the surface. The shallow band ($z > -3$ km) has μ_{app} median 0.277 with σ_n only ~23 MPa, so the stress-tied rule there gives $\mu_d \sim 0.13$ and small absolute drops — not a propagation bottleneck. No special free-surface μ_d treatment is required; the user's " $\mu_d \sim 0.1$ near the free surface" is already the fault-median outcome of Section 4.2.

5. Nucleation and friction parameters (hypocenter worked calc)

Worked nucleation/friction sheet for the SE-end hypocenter, updated from the V2 draft to (a) the C1 $k=2.39$ field facts and (b) the stress-tied friction of Section 4. Corrections vs the V2 draft are flagged inline. ($\delta_s=0.05$, $\delta_d=0.15$, $d_c=2.0$ m are the Section 4.2 / 8 proposed values, pending confirmation.)

5.1 Facts within the hypocenter (V3, C1 $k=2.39$ field)

Nucleation location (606971, 3707270, -4965.62); nearest DR facet centroid (607030, 3707289, -5054 m), 108 m away — in the SE region where $\mu_{app} \sim 0.30-0.35$ (the easiest place to nucleate). Stress from the C1 projection summary; material from `toolbox/hypocenter_facts/hypocenter_facts.py`:

Effective normal stress	$\sigma_N = 70.1 \text{ MPa}$	(was 63.8: P_p is now HYDROSTATIC $\sim 49.6 \text{ MPa}$ at 5 km, not 20)
Shear stress magnitude	$\tau_0 = 20.8 \text{ MPa}$	(was 21.2)
Apparent friction	$\mu_{app} = 0.296$	(was 0.332)
Shear-wave speed	$V_s = 3032 \text{ m/s}$	(MUSCAL; SeisSol-linear 3002)
Density	$\rho = 2551 \text{ kg/m}^3$	(SeisSol-linear 2541)
Shear modulus	$\mu = 23.5 \text{ GPa}$	(SeisSol-linear 22.9)
Poisson ratio	$\nu = 0.275$	(SeisSol-linear 0.278)

Material is UNCHANGED from V2 (same MUSCAL point); only the stress changed, because V3 adopts the C1 field, not V2's constant regional tensor. Note the C1 hypocenter traction is more strike-parallel ($\tau_{strike} 19.8$, $\tau_{dip} 6.4 \rightarrow$ rake $\sim +18$ deg) than V2's (rake -58 deg) — favourable for a right-lateral strike-slip start.

5.2 Linear slip-weakening friction (stress-tied; Section 4)

CORRECTION vs the V2 draft: a UNIFORM $\mu_s = 0.43$ is INFEASIBLE for the C1 field — the fault-wide max μ_{app} is 0.449 (not 0.42), so the 4.3 % of facets with $\mu_{app} > 0.40$ would PRE-SLIP at $t=0$. Requiring μ_s above the overall apparent friction therefore forces $\mu_s \geq 0.45$ if uniform, which then cannot rupture the $s=40\text{--}60 \text{ km}$ gate (Section 4.1). V3 ties friction to the local μ_{app} , so AT THE HYPOCENTER:

```
mu_s = mu_app + delta_s      = 0.296 + 0.05 = 0.346    (> mu_app: no pre-slip)
mu_d = max(mu_app - delta_d, 0) = 0.296 - 0.15 = 0.146    (fault median ~0.125)
(mu_s - mu_d) = 0.20         d_c = 2.0 m
```

Process zone (mesh must resolve it with 4-5 elements):

```
Lb = mu*d_c / ((mu_s-mu_d)*sigma_N) = 23.5e9*2.0 / (0.20*70.1e6) = 3352 m
mesh edge dx ~ 430-500 m (500 m mesh) -> Lb/dx ~ 6.7-7.8 elements OK
CORRECTION: the V2 draft's "Lb=2138 > 4*dx=2500" was false (2138<2500);
here Lb=3352 exceeds 4*dx for any dx in [430,625] m.
Deeper facets shrink Lb (sigma_N grows): Lb=1958 m at the gate
(sigma_N=120), 783 m near the deep barrier (sigma_N=300) -> marginal
there only (Section 4.3).
```

Static-friction sanity (no pre-slip):

```
mu_s = 0.346 > mu_app = 0.296 at the hypocenter, AND mu_s = mu_app + 0.05
everywhere by construction -> the whole fault is below failure at t=0.
```

Strength excess:

```
SE = tau_s - tau_0 = (mu_s - mu_app)*sigma_N = 0.05*70.1e6 = 3.5 MPa
(was 6.2 MPa; smaller by design -> nucleation is EASIER under V3.)
```

Nucleation length (Uenishi-Rice 2003; the patch must cover it):

```
Lnuc = 1.158*mu*d_c / ((1-nu)*(mu_s-mu_d)*sigma_N)
      = 1.158*23.5e9*2.0 / (0.725*0.20*70.1e6) = 5355 m    (half-length 2677 m)
CORRECTION: the draft's "Lnuc = mu*d_c/(...)" header was the PROCESS-ZONE
formula; the nucleation length is the U-R form above (with 1.158/(1-nu)).
```

Nucleation patch (gradual over-stress: time-ramped Gaussian T_{nuc_s} , $T_{nuc}=1$ s):

```
R = 6000 m    (e-fold radius; patch diameter 2R = 12000 m > Lnuc 5355 m)
dtau_nuc(min) = 1.1*tau_s - tau_0 = 1.1*0.346*70.1 - 20.8 = 5.9 MPa
    choose dtau_nuc = 10 MPa    (peak = tau_0 + 10 = 1.27*tau_s; robust margin)
over-stressed footprint r_os = R*sqrt(ln(dtau_nuc/SE))
    = 6000*sqrt(ln(10/3.5)) = 6143 m > Lnuc/2 = 2677 m    (onset ratio 2.3)
CORRECTION: the draft's r_os line mixed R=3500/5500/6000 and SE=6.24;
with R=6000 and SE=3.5 MPa -> r_os=6143 m. Even dtau=6 MPa gives
r_os=4399 m > Lnuc/2, so nucleation is robust; 10 MPa keeps V2's margin.
```

Bottom line: under V3 the SE-end hypocenter nucleates MORE easily than V2 (smaller strength excess, larger over-stressed footprint), so $R=6000$ m + $d\tau=10$ MPa is ample. The hard problem is PROPAGATION through the $s=40$ -60 km gate (Section 4.1), which the stress-tied μ_s addresses — not nucleation. Re-run `hypocenter_facts.py --case-dir safs_seis01_v3_0_0_LSW` once the V3 stress/friction nc exist to confirm these facts on the actual case inputs.

6. Implementation plan (files, once Section 8 is decided)

Phase 1 - stress converter + nc: - create `toolbox/.../csm_stress_to_asagi.py` (Section 3.1), run it -> `safs_seis01_v3_0_0_LSW/safs_stress_csm.nc`; converter self-check passes. Phase 2 - friction fields: - extend the converter (or a sibling) to emit `safs_friction_csm.nc` (μ_s , μ_d per Section 4.2, rasterized per 4.4); self-check vs VTU. Phase 3 - case files (copy V2, edit): - `parameters.par` (V2 verbatim except `EndTime`; Section 8 decision 5), - `safs_initial_stress.yaml` -> `!ASAGI safs_stress_csm.nc` (Section 3.2), - `safs_fault.yaml` -> `[mu_s]/[mu_d] !ASAGI safs_friction_csm.nc + deep barrier Lua + d_c/cohesion ConstantMap + Tnuc_s Lua` (Section 5), - reuse `safs_material_cvm.yaml` + `.nc` + `safs_mesh.puml.h5`, - `README_port.md` (run guide + the validation gates of Section 7).

7. Validation gates (cluster; cannot be checked locally)

- ☐ Converter self-checks pass (stress nc and friction nc reproduce the projection VTU per-facet values within tolerance) — LOCAL, must pass before submitting.
- ☐ $t=0$ stability: at $t=0$ NO facet is at failure ($\max \mu_{app} < \min \text{local } \mu_s$ by construction = δ_s); confirm from the first step's fault output $Ts0/Td0/Pn0$ vs the projection VTU (should match to interp).
- ☐ Nucleation: slip initiates at the hypocenter patch and the cohesive zone is resolved ($\Lambda \geq \sim 3$ facets except the deep barrier band).
- ☐ GATE CROSSING: the rupture front passes $s=40$ -60 km (the San Gorgonio gate) and $s=220$ -

240 km without arrest (rupture-time RT field continuous across both); this is THE acceptance criterion.

- ☐ Full propagation SE ($s \sim -18$ km) $\rightarrow$ NW ($s \sim +266$ km); moment does not plateau before the NW end (the failure signature of V2 B/C).
 - ☐ EndTime margin: at ≥ 3 km/s the ~ 285 km length needs ~ 95 s; set EndTime with margin (Section 8 decision 5).
-

8. Open decisions (confirm before the build)

1. Friction model = stress-tied μ_s/μ_d (Section 4.2). CONFIRM, and the values: $\delta_s = 0.05$, $\delta_d = 0.15$, $\mu_{d_floor} = 0.00$ ($S = 0.33$; median $\mu_d \sim 0.125$). Tighter S (smaller δ_d) = stronger drops / lower μ_d ; looser δ_s = safer $t=0$ margin but harder gates.
 2. Stress mechanism: (A) volumetric tensor ASAGI + SeisSol projects (clean, but SeisSol-recomputed μ_{app} differs from the tool's by interpolation, so the δ_s margin is approximate), vs (B) rasterized per-facet $Ts0/Td0/Pn0$ (exact margin, same machinery as the friction nc). RECOMMEND B for an exact, guaranteed-stable $t=0$ margin.
 3. Friction injection mechanism: Option A (rasterized ASAGI) vs verifying a native per-facet path in the QuakeWorx SeisSol build (Section 4.4).
 4. $d_c = 2.0$ m uniform vs d_c tied to σ_n for constant Λ (Section 4.3) — affects only the deep band.
 5. EndTime (parameters.par): 150-200 s (V2 used 200 s) given ~ 285 km.
 6. Grid resolution of the stress/friction nc : $dx=dy=500$ m vs 250 m near the fault junctions (Section 4.4 junction risk).
-

9. References (SAF-system / mechanics; verification status as of 2026-06-15)

- Lachenbruch, A.H. and Sass, J.H. (1980, 1992). Heat-flow bound on SAF-resolved shear ($\sim \leq 20$ MPa average) $\rightarrow$ low apparent/dynamic friction.
- Hardebeck, J.L. and Hauksson, E. (2001), JGR 106, 21859-21882. Southern California: weak SAF in a low-strength crust; crustal deviatoric stress ~ 10 MPa (Landers stress rotation) — supports low μ_d .
- Hardebeck, J.L. and Michael, A.J. (2004), JGR 109, B11303. Intermediate- strength SAF ($\mu \sim$ half a strong fault), the basis for $k=2.39 / \mu \sim 0.45$ in the stress field this case adopts.
- Townend, J. and Zoback, M.D. (2004), GRL 31, L15S11; Jones (1988), JGR 93, 8869. $SH_{max} \sim 65$ deg to the southern SAF $\rightarrow$ weak fault, low resolved friction.
- The LSW cohesive-zone-length estimate Λ (Section 4.3) is the same standard ($9\pi/32$) form V2 used (Day-type / Palmer-Rice; e.g. Day, S.M. et al. 2005, JGR 110, B12307) — setting-independent mechanics; verify the exact citation/pages before any manuscript use.
- Uenishi-Rice nucleation half-length (Section 5; Uenishi and Rice 2003, JGR 108(B1), 2042) — the same nucleation criterion V2 used; setting-independent mechanics; verify pages before manuscript use.